DERIVED PULLBACK PRESERVES QUASI-COHERENCE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 627-642, immediately following the statement at 619-625.

Editorial note. The commutative square with surjective a is the span's own setup; scheme case is the terminal input.

Allowed standard prerequisites (editorial).

- The covering criterion for quasi-coherence. Lemma lemma-check-quasi-coherence-on-covering of this chapter, cited at proof lines 637-641. Reduces the claim to the scheme case once the pullback commutes with the etale diagram.
- Scheme case. Derived Categories of Schemes, Lemma perfect-lemma-quasi-coherence-pullback, cited at proof lines 638-641. Terminal input: Lh* preserves D_QCoh for schemes.

Source reference. Original lemma 08F2: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 535–535. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08F4: focused-seeds/geometry/S9-perf-quasi-coherence-pullback/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : Y \to X$ be a morphism of algebraic spaces over S. The functor Lf^* sends $D_{QCoh}(\mathcal{O}_X)$ into $D_{QCoh}(\mathcal{O}_Y)$.*

Proof. Choose a diagram

$$\begin{array}{ccc} U & \xrightarrow{h} & V \\ {\scriptstyle a}\downarrow & & \downarrow{\scriptstyle b} \\ X & \xrightarrow{f} & Y \end{array}$$

where U and V are schemes, the vertical arrows are étale, and a is surjective. Since $a^* \circ Lf^* = Lh^* \circ b^*$ the result follows from Lemma 08F2 and the case of schemes which is Derived Categories of Schemes, Lemma 08F4. □

Date: stacks-project:spaces-perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.